Formula 1 Race Launch with State-dependent Phase Optimisation

Formula 1 Race Launch with State-dependent Phase Optimisation

Marc-Philippe Neumann[1], Matteo Babin[1], Giona Fieni[1], Oliviero Agnelli[3], Armin Nurkanović[2], Alberto Cerofolini[3], Christopher H. Onder[1]

***Abstract*—The launch of a Formula 1 race represents a crucial stage that influences the result. It is characterised by multiple consecutive state-dependent phases. At the starting signal, the drivers gradually release the clutch paddle. Then, with a locked clutch, the gear selection determines the subsequent phases. Each phase evolves according to distinct dynamics, while accepting different control inputs. In this work, we propose a framework that jointly optimises those phases. Specifically, we optimise the phase-specific control inputs and the switching times. Results show that decreasing the available battery energy by 0.1 MJ affects the gear shifting strategy and increases the time to cover 140 m by 6 ms. This result validates the superiority over sequential phase optimisation, which potentially leads to infeasible results due to its causal nature. For a wet track scenario, we show the duration increase of the phases, while complying with optimal torque deployment for best acceleration.**

***Index Terms*—Nonlinear Optimal Control; Hybrid Dynamical System; Friction Clutch Optimisation; Formula 1; Race Launch Optimisation; State-dependent Phase Optimisation; Hybrid Electric Vehicles.**

I. Introduction

FORMULA 1 (F1) has excited motor sport fans all over the world since 1950. Since then, the greatest drivers of four-wheeled racing vehicles have been competing for the prestige of holding the title of this championship. Although the change to a hybrid powertrain in 2014 has been controversially discussed among the public, it has brought great engineering potential to the competition. In fact, today's race cars are equipped with a turbocharged 1.6 L V6 internal combustion engine (ICE), electrically assisted by a motor generator unit – kinetic (MGU-K). Further, the turbocharger is electrically assisted by a motor generator unit – heat (MGU-H). This component allows to recuperate excess energy produced by the turbine, or boost the turbocharger during transient operation avoiding turbo-lag phenomena. In the following referred to as the power unit (PU), this propulsion system is linked to the wheels via an eight-speed sequential gearbox and a limited-slip differential. Fig. 1 shows the powertrain model schematically. The optimal synergy of all these components is crucial, such that from an engineering perspective, the vehicle is competitive

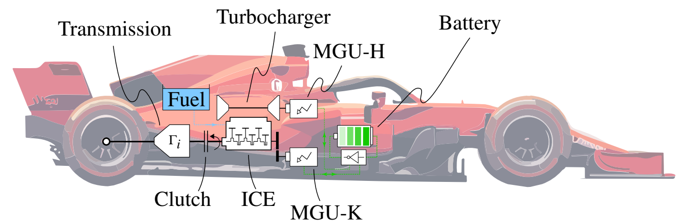

Figure 1: Formula 1 (F1) car with a schematic representation of the powertrain. From left to right, there is the transmission with a gear ratio Γ_i related to the i-th gear. Then, there is a friction clutch, a 1.6 L V6 internal combustion engine (ICE) with its electrically assisted (MGU-H) turbocharger, and an electric motor (MGU-K). In blue and green, we depict the energy reservoirs, i.e., the fuel tank and the battery, respectively.

and able to fit the driver's requirements in order to win races. However, in order to level the playing field, each season the Fédération Internationale de l'Automobile (FIA) issues an updated version of detailed regulations to be complied with [1], [2].

Each Grand Prix (GP) consists of a sequence of laps, resulting in a race distance of about 300 km. If we neglect the influence of strategic factors such as pit-stops, overtakes, or the deployment of the safety car, as well as tyre wear considerations and weight decrease, each lap is very similar. Only one phase of the race is structurally different: the launch. After an initial formation lap, where the drivers complete one lap at reduced speed to bring each component to optimal temperature, each car is stopped in a predefined box on the starting grid. Once the last car is standing still, the five red lights turn on sequentially, and the drivers increase the engine speed of their cars. Once all five lights are on, they suddenly and randomly go out. After an average reaction time of only 0.2 s [3], the wheels start turning. Similar to a regular car with manual transmission, in F1 the driver must engage the clutch paddle on the steering wheel in order to synchronise the engine speed and the layshaft speed. After a short period of clutch *slipping*, the paddle is fully released and the clutch is called *locked*. From this point on, the torque modulation is achieved by means of the throttle pedal. Additionally, a high-performance semi-automatic sequential gearbox with paddle-shifters is actuated whenever the driver requests a gear shift. The focus of each driver is now to reach the first braking zone in the quickest time possible, while avoiding contact with other competitors. In Fig. 2 we show the averaged overtake data, obtained by comparing the grid standings at the start, at the

[1]Institute for Dynamic Systems and Control, ETH Zürich, Zürich, Switzerland, {mneumann,mbabin,gfieni,onder}@ethz.ch
[2]Department of Microsystems Engineering (IMTEK), University of Freiburg, Freiburg, Germany, armin.nurkanovic@imtek.uni-freiburg.de
[3]Power Unit Performance and Control Group, Ferrari S.p.A., 41053 Maranello, Italy, Alberto.Cerofolini@ferrari.com

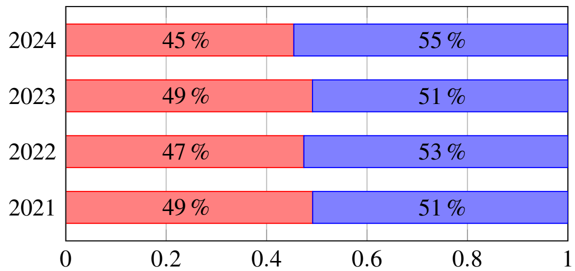

Figure 2: Averaged overtake data for the past four F1 seasons. Specifically, for each season from 2021 we show the average percentage of overtakes that occurred in the first lap in red, and the average percentage of overtakes during the rest of the race in blue. To obtain the number of overtakes, we compare the grid positions at laps 1, 2, and the end, respectively. It can be seen how, although the horizon differs from one single lap to over 40, the number of overtakes is very similar. This highlights the importance of the first lap on the race outcome.

end of the first lap, and at the end of the race for all races of the past four seasons (i.e., 2021 – 2024) [4]. It can be clearly seen how the percentage of overtakes occurring only during the first lap is very close to the one happening during the rest of the race. This highlights the importance of a successful first lap, which must be initiated with an optimal launch of the race.

A. *Related work*

Given the problem setting of race launch optimisation, in which we consider sudden changes in the system's dynamics, we subdivide the investigated literature into three streams.

The first one considers the modelling of the components of interest. The overall dynamics of a vehicle can be modelled via the fundamental principles of car physics [5]. For racing vehicles, additional focus is put on aerodynamic drag and downforce, tyres, and loads, as described in [6], [7]. In [8] a comprehensive approach to vehicle dynamics is presented, featuring a non-rigid vehicle model with 14 degrees of freedom (DOF) for longitudinal, lateral, and vertical directions. Given the high complexity of many models, a more mainstream approach is provided by [9], where rigid, 3 DOF models provide sufficient accuracy for lap time simulation. For the purpose of optimisation, simpler modelling approaches are essential. In light of the fact that we consider only the launch phase of a race, we can neglect lateral influences and focus on longitudinal dynamics only, reducing the car model to one spatial DOF. In consequence, we simplify the principles of generic longitudinal models given by [10], [11]. The primary challenge posed by the race launch problem setting is the modelling of the clutch, mounted on the drive shaft. While it transitions from a disengaged status, through a slipping phase up to fully engaged locking condition, the overall drivetrain dynamics change structurally. In the literature, we find a unified formulation for these phases, such as the Karnopp approach [12], [13]. However, in this article we propose a model, where the drivetrain dynamics switch, based on the clutch state [14]. For the slipping phase of the launch, an accurate clutch model is crucial. F1 clutches are highly complex systems [15]. Their dynamics include high order thermal characteristics [16]–[18] and stick-slip behaviour [19]–[21]. It is known that the friction coefficient is dependent, among other things, on the slip speed at the clutch friction interfaces [22]. To date, clutch engagement optimisation has primarily focused on road cars, where the objective is to achieve smooth engagement for comfortable operation [23], [24]. An additional topic of interest focuses on the tyres, as they represent the only contact point between the car and the track. More specifically, the tyre model determines how much of the torque produced by the power unit is in fact transferred to the track, and thus contributes to vehicle propulsion, resulting in a central component for race launch optimisation. A well-known modelling approach is the brush tyre model [25], [26]. Alternatively, a less physical approach based on a fully parametric relationship, is the Pacejka tyre model [27], [28]. Both approaches rely on the slip ratio, which poses a problem for low-speed calculations such as the race launch. To address this issue, most literature adopts a switch technique [29]. Given the scientific acceptance of the Pacejka tyre model, in this article we employ an adapted version of it, with a reduced number of parameters and a slip speed rather than slip ratio dependency.

The second stream of research addresses optimising the race launch strategy itself. In road vehicles, fuel optimal control strategies were investigated by means of dynamic programming (DP) [30]–[32], convex optimisation [33], [34], and Pontryagin's Minimum Principle (PMP) [35]–[37]. Similarly, [33] introduces a method based on deterministic DP and convex optimisation for energy management in hybrid electric vehicles, incorporating costs for engine starts and gear shifts. Further, integer variables such as gear shifts and engine deactivation were considered and optimised by means of iterative linear programming [38], via PMP [39] and mixed-integer linear programming (MILP) [40]. In F1, the main goal is to minimise lap time while complying with the regulations [1], [2]. For example, [41] proposes a method using quasi-steady-state models to solve this problem, while also optimising the trajectory. An optimal energy recovery system control was analysed in [42], while considering a less detailed ICE model. Furthermore, studies [43], [44] have addressed the time-optimal control problem in a computationally efficient manner, enabling real-time application. In [45], the authors present a detailed low-level mathematical model of the F1 powertrain suited for numerical optimisation. The insights of that article have been further included in online control strategies in [46].

Finally, the third stream of research involves the optimisation methods. One of the primary challenges is managing the switch between an unlocked and locked clutch, without incurring the computational burden of mixed-integer optimisation or non-smoothness. Studies [47], [48] present the time-freezing reformulation to handle state jumps, where a clock state is introduced that does not evolve during the runtime of the auxiliary system. Another approach follows the principle of numerical optimal control with switch detection of rigid bodies with impacts and Coulomb friction [49]. However, the main feature of that paper is that the order and number of phase changes are not known a priori, which makes the approach more general but also computationally demanding. Finally,

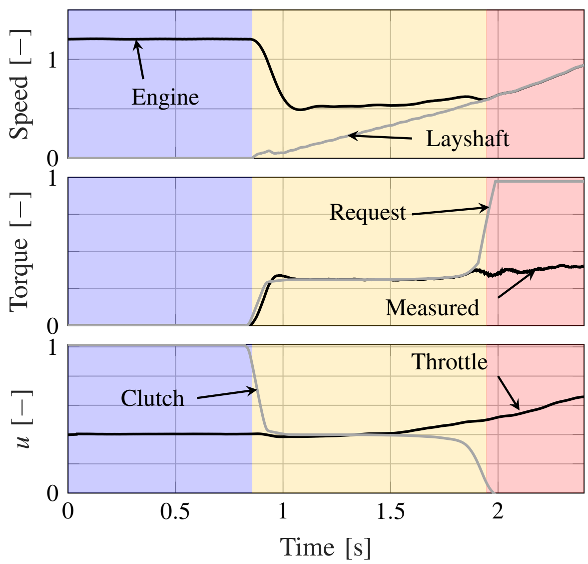

Figure 3: Artificial trajectories of the primary signals of interest, replicating a launch scenario. In the top subplot we show the engine and layshaft speeds (clutch input and output, respectively). The middle subplot depicts the clutch torque request and the measured torque. Finally, the bottom subplot displays the main inputs at the driver's disposal. The coloured patches represent the different phases of our system. In blue we show Phase 0, where the vehicle is standing on the starting grid. In yellow we depict Phase 1, where the clutch is slipping. Finally, in red, we display Phase 2, where the clutch is locked.

[50] proposes a time-freezing reformulation method for hybrid systems with hysteresis.

In summary, the literature review yields major gaps in the optimisation of race launch scenarios. From an optimisation point of view, clutch modelling is usually performed by combining its distinct dynamics with a single continuous relation, leading to inconsistent results. As a consequence, the crucial launch phase, where state-dependent changes in the dynamics occur, is not considered. In particular, state-of-the-art optimisation frameworks consider optimal laps without starting from standstill (so called flying laps). Mathematical optimisation methods considering state-switches and hybrid systems offer efficient techniques to bridge this gap.

B. Problem Statement

The dynamics of a race launch represent a complex series of events, whose order of occurrence is known a priori. In Fig. 3 we show the main signals that are subject to analysis in this work: shaft speeds, clutch torque trajectories, and control inputs. We use a colour scheme to depict three different phases: standstill in blue, clutch slip in yellow, and clutch lock in red. Starting from a vehicle at standstill, the driver initiates the acceleration by operating the clutch paddle and the throttle pedal. In this phase, the engine and layshaft speeds (clutch input and output, respectively) differ. Once the clutch is fully engaged, the system structurally changes in two ways. First, the clutch input ceases to be available, while the decision variable for the sequential gear shifts is now included to optimally accelerate the race car up to the first corner. Second, the drivetrain model switches from two rotational DOF, to a single DOF. The torque trajectories show how they match as long as the clutch is not fully engaged. Once the clutch is locked, the torque request always exceeds the maximum transferable torque.

In this paper we will refer to those phases as follows:

- **Phase 0** represents the interval in which the car is not moving. During this phase, the engine speed is usually kept at an optimal and reliable level for the race launch by actuating the throttle pedal while keeping a fully disengaged clutch.
- **Phase 1** begins as soon as the driver starts releasing the clutch paddle. During this phase the clutch is slipping (i.e., the clutch input and output shafts do not rotate at the same speed), and we assume to be in first gear.
- **Phase 2** begins when the clutch paddle is fully released and therefore the clutch is locked. From that point on, the driver controls the torque at the wheels solely through the throttle pedal. Given the ability to shift gears, we further divide Phase 2 into consecutive sub phases, each representing a certain engaged gear.

The primary focus of this article comprises only Phases 1 and 2, as the high-level model used does not contemplate low-level PU dynamics relevant in Phase 0.

C. Contribution

There are two key contributions that underscore the relevance of this work. First, we develop a framework for the race launch optimisation by incorporating varying state-dependent dynamics. To the best of the author's knowledge, for the first time a clutch locking and unlocking behaviour is jointly optimised in a race launch scenario. Specifically, we aim to answer the following questions:

> *What is the optimal clutch actuation to guarantee the quickest acceleration?*
> *How does the clutch actuation and torque deployment vary, depending on energy budget targets or track conditions?*

Second, we improve the computational tractability of the nonlinear and nonconvex optimal control problem (OCP) formulation at hand, by exploiting the known phase order of the hybrid system. We apply a state-of-the-art method of optimal control with phase switches, in order to also optimise the time instants where the phases change. The questions we address are the following:

> *Do the times spent in the various phases vary depending on optimisation targets?*
> *If yes, how are the optimal trajectories influenced by such duration variations?*

D. Paper Structure

In Section II we introduce the models employed in our framework. In Section III, we then describe the methodology adopted in this paper. We first familiarise the reader with some key concepts of hybrid systems and draw the boundaries of each phase. Next, we formulate the entire OCP. By means of two case studies emulating real life scenarios, we investigate the robustness and modularity of our framework in Section IV.

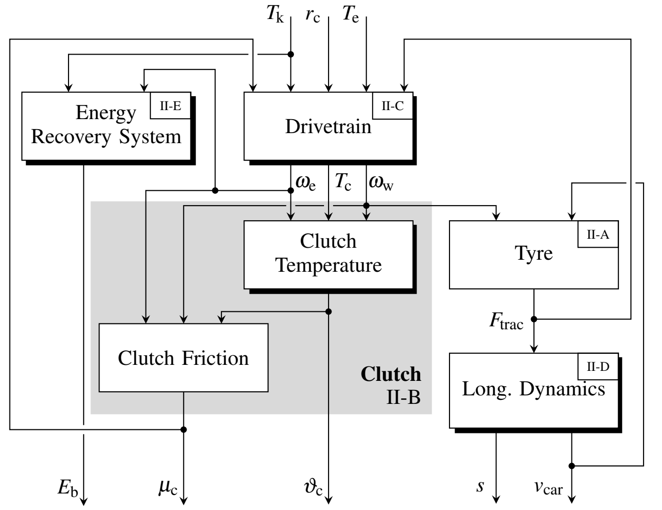

Figure 4: Cause and effect diagram of the F1 system that we consider in the optimisation framework presented in this paper. The blocks with a black shading represent dynamic systems, i.e., where a differential equation is at the core. The other blocks represent quasi-static subsystems, characterised by analytical relations. To facilitate the interpretation, we include the corresponding section.

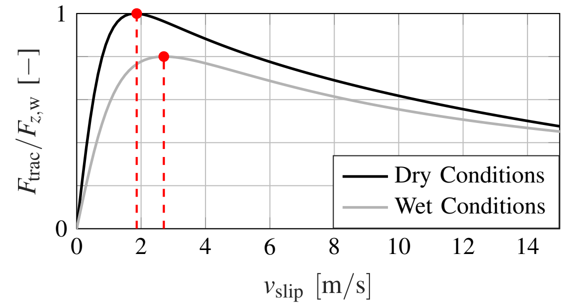

Figure 5: Normalised tyre traction force as a function of the slip speed. In black we show the curve for a dry track scenario, while in grey for wet track conditions. For both scenarios, the dashed vertical lines indicate the optimal slip speed value $v_{slip,opt}$ that achieves the highest possible traction force F_{trac}.

Furthermore, we propose a qualitative comparison of the joint optimisation to the single phase optimisation, to assess the optimality and necessity of the former. Finally, in Section V we draw conclusions, comment on the relevant insights gained through our framework, and propose an outlook for future research.

II. F1 Dynamic Model

This section introduces the model of the F1 vehicle considered in this paper. In Fig. 4 we display the cause and effect diagram of the system. It indicates the dependencies between subsystems by depicting the signal flows. To facilitate the interpretation, we also include a reference to the corresponding section.

A. Tyre

In this section we introduce the first model of this research work, concerning the tyres. As previously mentioned, they represent a crucial component in racing, as they determine how much of the torque that the PU delivers can actually be transferred to the track, and thus contributes to vehicle propulsion. The tyre is the only point of contact between the racing car and the track. Given the immense torque deployment achievable by such PUs, an accurate tyre modelling is crucial for obtaining sensible race launch trajectories. An important quantity that determines the magnitude of transmitted force in most tyre models available in the literature is the tyre's longitudinal slip ratio r_{slip}, which under acceleration is defined as

$$r_{slip}(t) = \frac{R_w \cdot \omega_w(t) - v_{car}(t)}{R_w \cdot \omega_w(t)}, \tag{1}$$

where R_w is the unloaded wheel's radius, ω_w is the wheel's rotational speed, and v_{car} is the vehicle's velocity. However, Equation (1) poses problems during standstill phases, where the wheel speed is zero, as well as during acceleration phases, where significant changes in relative speed cause high magnitude fluctuations. Both these behaviours are highly undesired in an optimisation framework. To cope with these issues, we introduce the longitudinal slip speed v_{slip}, defined as the difference between the scaled wheel's rotational speed and the actual ground speed of the vehicle. In the acceleration case that we consider in our work, it is defined as

$$v_{slip}(t) = \omega_w(t) \cdot R_w - v_{car}(t).$$

To model the propulsive force transmitted through the tyres, we adopt a model based on the longitudinal Pacejka tyre formula [27], [28], reformulating it with a reduced set of parameters:

$$F_{trac}(t) = F_{z,w}(t) \cdot \alpha_{t,1} \cdot \sin(\Psi(t)), \tag{2}$$

where $F_{z,w}$ is the normal force applied on a single rear tyre, Ψ is defined as

$$\Psi(t) = \alpha_{t,2} \cdot \arctan\Big(\alpha_{t,3} \cdot (1-\alpha_{t,4}) \cdot \frac{v_{slip}(t)}{v_{ref}} + \dots \\ \alpha_{t,4} \cdot \arctan(\alpha_{t,3} \cdot \frac{v_{slip}(t)}{v_{ref}})\Big),$$

and the model parameters $\alpha_{t,1}$, $\alpha_{t,2}$, $\alpha_{t,3}$, $\alpha_{t,4}$, and v_{ref} are subject to identification. To identify the parameters, we analysed telemetry data from previous race launches and extracted key performance indicators (KPIs), such as the time from 0 to 100 km/h, making sure we are able to replicate them. While this model is significantly simpler compared to more complex multidimensional models, it is able to capture the behaviours that are relevant during a race launch without compromising the optimisation. We assume the optimal slip speed to be constant, and that overspinning the tyres leads to a reduction in traction force. Fig. 5 displays the resulting normalised traction force of the model as a function of the tyre slip speed. We present this model for two distinct scenarios, namely, in black for dry track conditions, and in grey for wet track conditions. There exists an optimal value for the tyre slip speed, i.e., $v_{slip,opt}$, which maximizes the longitudinal friction coefficient and thus the traction force. We refer to this maximum traction force as $F_{trac,max}$.

The normal force in Equation (2) is expressed as the sum of three distinct components

$$F_{z,\mathrm{w}}(t) = F_{z,\mathrm{static}} + F_{z,\mathrm{transfer}}(t) + F_{z,\mathrm{aero}}(t), \tag{3}$$

where $F_{z,\mathrm{static}}$ is a constant force due to static weight (we assume constant mass for the short time horizon considered), $F_{z,\mathrm{transfer}}$ represents the additional force stemming from the longitudinal weight transfer during acceleration, and $F_{z,\mathrm{aero}}$ is the component stemming from aerodynamic downforce. Note that we describe each force for a single rear tyre, as the vehicle is rear wheel drive. The first component is defined as

$$F_{z,\mathrm{static}} = \frac{m_{\mathrm{car}} \cdot g \cdot \mathrm{wb}_{\mathrm{rear}}}{2}, \tag{4}$$

where m_{car} is the car's mass, g is the gravitational acceleration, and $\mathrm{wb}_{\mathrm{rear}}$ is the rear car weight balance. The variable transfer component reads

$$F_{z,\mathrm{transfer}}(t) = \frac{k_{\mathrm{d}} \cdot m_{\mathrm{car}} \cdot a_{\mathrm{car}}(t) \cdot \frac{h_{\mathrm{cog}}}{L_{\mathrm{wb}}}}{2}, \tag{5}$$

where k_{d} is a constant damping factor which accounts for the rigidity of the rear suspension that compresses during the vehicle's acceleration a_{car} and reduces the weight transfer, h_{cog} is the height of the centre of gravity, and L_{wb} is the car's wheelbase. Finally, the aerodynamic component reads

$$F_{z,\mathrm{aero}}(t) = \frac{\frac{1}{2} \cdot c_{\mathrm{l}} \cdot A_{\mathrm{car}} \cdot \rho_{\mathrm{air}} \cdot v_{\mathrm{car}}(t)^2 \cdot \mathrm{ab}_{\mathrm{rear}}}{2}, \tag{6}$$

with A_{car} being the car frontal area, ρ_{air} the air density, $\mathrm{ab}_{\mathrm{rear}}$ the aerodynamic balance to the rear of the car, and c_{l} the aerodynamic lift coefficient.

B. Clutch

The clutch is a friction component that is used to synchronise the rotational speeds of two shafts connected to it. In F1 race cars, the clutch is mounted between the power unit and the rest of the drivetrain (gearbox, differential, wheels). The shaft on the power unit's side is called engine shaft, while the one downstream is called layshaft. Whenever the vehicle is operated below a certain velocity, the employment of the clutch becomes mandatory in order to avoid the ICE to stall. Specifically, when the rotational speed of the layshaft is lower than the idle speed of the engine. In a race launch scenario, this situation starts when the vehicle is standing still, and lasts up to the moment when the engine and the layshaft speeds are matched. In our framework, we assume that the race launch is started in first gear and that clutch slipping occurs exclusively then. This assumption was made based on the analysis of telemetry data during dry weather conditions. The main command that is relevant for a successful synchronisation is the clutch paddle position r_{c},

$$0 \leqslant r_{\mathrm{c}}(t) \leqslant 1. \tag{7}$$

The convention used in this paper demands that $r_{\mathrm{c}} = 1$ corresponds to the clutch paddle being fully pulled, thus the clutch being fully disengaged. Consequently, $r_{\mathrm{c}} = 0$ indicates the clutch paddle is fully released and the clutch is locked. It is the competence of the driver to engage this component in an optimal way, as traction control and other supportive measures are not allowed by the regulations [1]. Clearly, there are many external factors that influence the effectiveness of the friction clutch. For example, the friction coefficient might be affected by several factors such as temperature, clutch wear, and environmental conditions. In this paper, we include a model to express the friction coefficient μ_{c} as a function of the rotational slip speed $\omega_{\mathrm{c,slip}}$ between the discs and plates, and the clutch temperature ϑ_{c} [22], i.e.,

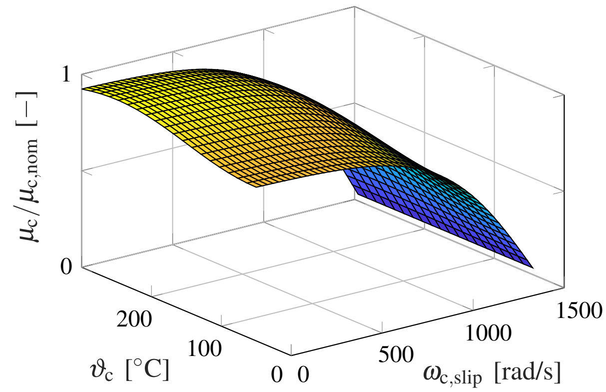

Figure 6: Graphical representation of the parametric model (8), highlighting the dependency of the clutch's friction coefficient on the temperature and the slip speed. It can be seen how the slip speed has a more significant influence than the temperature.

$$\mu_{\mathrm{c}}(t) = \mu_{\mathrm{s}} + (\mu_{\mathrm{t}}(t) - \mu_{\mathrm{s}}) \cdot \tanh(\beta(t)), \tag{8}$$

where μ_{s} is the static friction coefficient, μ_t is the temperature-dependent part of the model

$$\mu_{\mathrm{t}}(t) = \alpha_{\mathrm{c},1} \cdot \sin(\alpha_{\mathrm{c},2} \cdot \vartheta_{\mathrm{c}}(t) + \alpha_{\mathrm{c},3}) + \alpha_{\mathrm{c},4},$$

and β is defined as

$$\beta(t) = \gamma \cdot \left(\alpha_{\mathrm{c},5} - \frac{30}{\pi} \cdot \omega_{\mathrm{c,slip}}(t)^{\alpha_{\mathrm{c},6}} \right).$$

The dimensionless parameters $\alpha_{\mathrm{c},1}$, $\alpha_{\mathrm{c},2}$, $\alpha_{\mathrm{c},3}$, $\alpha_{\mathrm{c},4}$, $\alpha_{\mathrm{c},5}$, $\alpha_{\mathrm{c},6}$, and γ are subject to identification. The resulting model is depicted in Fig. 6, illustrating that the slip speed dependency has a more significant influence compared to the temperature dependency. The rotational slip speed reads as

$$\omega_{\mathrm{c,slip}}(t) = \omega_{\mathrm{e}}(t) - \frac{\omega_{\mathrm{w}}(t)}{\Gamma_1},$$

where ω_{e} and ω_{w} are the engine and wheel speeds, respectively. Note that we account for the gear ratio Γ_1 in first gear only. To compute the temperature dynamics, we model the clutch as a bulk system, incorporating a heat generation and a heat dissipation term, i.e.,

$$\frac{\mathrm{d}}{\mathrm{d}t}\vartheta_{\mathrm{c}}(t) = \frac{1}{c_{p,\mathrm{c}} \cdot m_{\mathrm{c}}} \cdot \left[P_{\mathrm{c}}(t) - c_{\mathrm{therm}} \cdot A_{\mathrm{surf}} \cdot (\vartheta_{\mathrm{c}}(t) - \vartheta_{\mathrm{amb}}) \right], \tag{9}$$

where $c_{p,\mathrm{c}}$ is the specific heat capacity of the clutch material, m_{c} is the mass of the clutch, c_{therm} is the convective heat transfer coefficient, A_{surf} is the surface area, and ϑ_{amb} represents the ambient temperature inside the clutch housing, assumed to be constant. The generated heat power is given by

$$P_{\mathrm{c}}(t) = \omega_{\mathrm{c,slip}}(t) \cdot T_{\mathrm{c},j}(t), \qquad j \in \{1,2\},$$

where $T_{c,j}$ is the torque transferred by the clutch during Phase j.

Phase 1. For the first time in this paper, at this point of the modelling we need to distinguish between the distinct phases of the launch. In particular, the clutch torque for Phase 1 reads

$$T_{c,1}(t) = \big(1 - r_c(t)\big) \cdot k_c \cdot \mu_c(t) \cdot R_c \cdot n_{int}, \tag{10}$$

where k_c is a gain used to replicate the normal force from the actuator, R_c is the clutch's equivalent friction radius, and n_{int} represents the number of friction interfaces. This implies that at disengaged clutch, the transferred torque $T_{c,1}$ is 0 Nm and thus no heat is generated, as $P_c = 0\,\mathrm{W}$.

Phase 2. In Phase 2, the torque transferred at the clutch $T_{c,2}$ is no longer dependent on the clutch paddle position, as the clutch is fully locked. Instead, it is the sum of torques downstream, depending on the engaged gear $i \in \{1, \ldots, n_{gears}\}$, going from first gear to the last available gear n_{gears}, i.e.,

$$T_{c,2}(t) = \frac{\Gamma_i(t)}{\eta_{dt}} \cdot \Big(2 \cdot R_w \cdot F_{trac}(t) + 2 \cdot \mu_w \cdot \omega_e(t) \cdot \Gamma_i(t)\Big), \tag{11}$$

where η_{dt} is a constant efficiency coefficient used to model gearbox friction losses, and μ_w is a constant friction coefficient, both introduced in Section II-C below. The maximum torque that the clutch could transmit before re-entering a non-favourable slipping state is higher than the actual transmitted torque. Additionally, in this model we neglect high frequency friction behaviour such as stick and slip dynamics. Being $\omega_{c,slip} = 0\,\mathrm{rad/s}$, no heat is generated as $P_c = 0\,\mathrm{W}$.

C. Drivetrain

In this paper, we model the drivetrain as a rotating mass system, where we combine the components downstream of the clutch as a single equivalent inertia. The gear ratios Γ_i represent the overall gear ratio from engine to wheel for each gear i. As anticipated in Section I-B, the drivetrain dynamics undergo two structurally distinct phases during the launch.

Phase 1. During Phase 1, when the clutch is slipping, the drivetrain can be modelled as a 2-DOF system. The layshaft speed ω_l is not constrained to be equal to the engine output shaft. The only requirement is that the torque on the input and output sides of the clutch must be equal, assuming the clutch introduces no additional rotational inertia. Additionally, the shafts are considered to be infinitely rigid. Fig. 7 shows the 2-DOF rotating mass model representing the drivetrain when the clutch is slipping. The angular momentum principle applied on the engine side results in

$$J_e \cdot \frac{d}{dt}\omega_e(t) = T_e(t) + T_k(t) - T_{c,1}(t) - T_{\mu,e}(t), \tag{12}$$

where J_e is the engine inertia, T_e is the engine torque, T_k is the MGU-K torque at the flywheel, and $T_{c,1}$ is the torque transferred at the clutch in Phase 1 according to Equation (10). The engine friction torque $T_{\mu,e}$ is modelled linearly depending on the engine's rotational speed

$$T_{\mu,e}(t) = \omega_e(t) \cdot \mu_e.$$

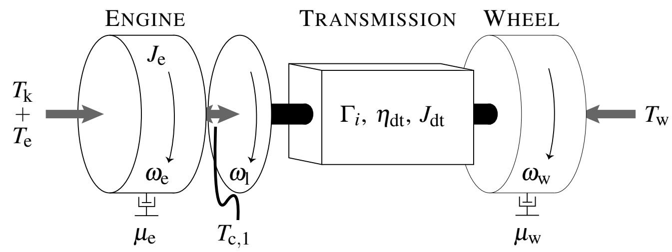

Figure 7: Model schematic of the 2-DOF system. The engine is modelled as a rotating mass with its inertia J_e and speed ω_e. The torques that are applied on this subsystem are the engine and MGU-K torques T_e and T_k, respectively, as well as the clutch torque $T_{c,1}$. Additionally, a speed-dependent friction torque is opposed to the movement with constant coefficient μ_e. On the other side of the clutch, a rotating mass with lumped inertia J_{dt} revolves at ω_l at the layshaft, and ω_w at the wheels, being strictly dependent on the engaged gear ratio Γ_i. There, the wheel torque T_w is applied in addition to a friction torque with coefficient μ_w.

The engine speed is constrained due to thermodynamic and mechanical limitations, i.e.,

$$\omega_{e,idle} \leqslant \omega_e(t) \leqslant \omega_{e,max}, \tag{13}$$

where $\omega_{e,idle}$ is the engine's idling speed, and $\omega_{e,max}$ is the maximum speed. For both the engine and the MGU-K torques we include torque-speed maps that constrain their magnitude, as shown in Fig. 8. The grey areas indicate the feasible torque-speed combinations. In particular, we can write

$$\begin{aligned} 0 \quad &\leqslant \quad T_e(t) \quad \leqslant \quad T_{e,max}(\omega_e(t)), \\ T_{k,min}(\omega_k(t)) \quad &\leqslant \quad T_k(t) \quad \leqslant \quad T_{k,max}(\omega_k(t)), \end{aligned} \tag{14}$$

where $T_{e,max}$ is the engine speed dependent maximum engine torque, and $T_{k,min} < 0$ and $T_{k,max}$ are the motor speed dependent minimum and maximum MGU-K torques, respectively. According to the regulations, below a car velocity of 50 km/h, the MGU-K torque is not allowed to be positive, resulting in an additional constraint

$$T_{k,min}(\omega_k(t)) \quad \leqslant \quad T_k(t) \quad \leqslant \quad 0, \quad \text{if } v_{car} < 50\,\mathrm{km/h}. \tag{15}$$

Downstream of the clutch, at the layshaft, the angular momentum principle leads to a second dynamic equation, i.e.,

$$J_{dt} \cdot \frac{d}{dt}\omega_w(t) = \frac{T_{c,1}(t) \cdot \eta_{dt}}{\Gamma_1(t)} - 2 \cdot \Big(R_w \cdot F_{trac}(t) + \mu_w \cdot \omega_w(t)\Big), \tag{16}$$

where J_{dt} represents the lumped inertias of all components in the drivetrain, from the clutch output shaft to the wheels, and the factor of 2 in front of the tyre torque and tyre friction terms accounts for both wheels.

Phase 2. In Phase 2, the clutch is fully locked and the synchronisation process of the layshaft and engine speeds is terminated, i.e.,

$$\omega_e(t) = \omega_l(t).$$

In consequence, the drivetrain now only has 1 DOF, with a single rotating mass model shown in Fig. 9. The differential equation for the engine's rotational speed is

$$\left(J_e + J_{dt} \cdot \frac{\Gamma_i(t)^2}{\eta_{dt}}\right) \cdot \frac{d}{dt}\omega_e(t) = T_e(t) + T_k(t) - T_{\mu,e}(t) - T_{c,2}(t). \tag{17}$$

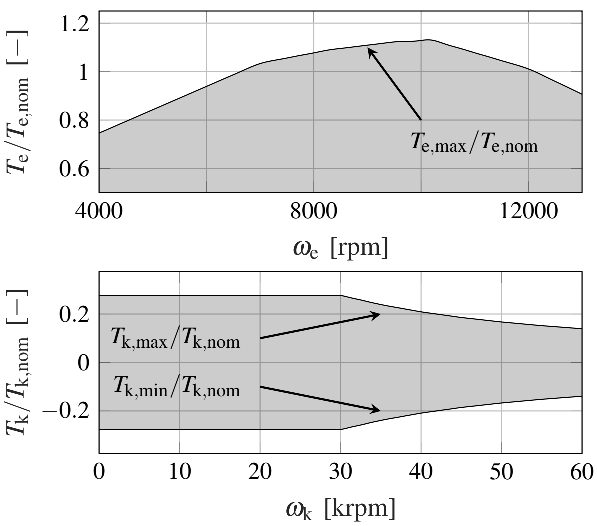

Figure 8: Normalised torque-speed maps for the engine torque T_e in the top plot, and the MGU-K torque T_k in the bottom plot. For both, we indicate the feasible region with a grey-patched area. Note that for the MGU-K there exists a negative torque constraint, representing the maximum recuperation torque allowed.

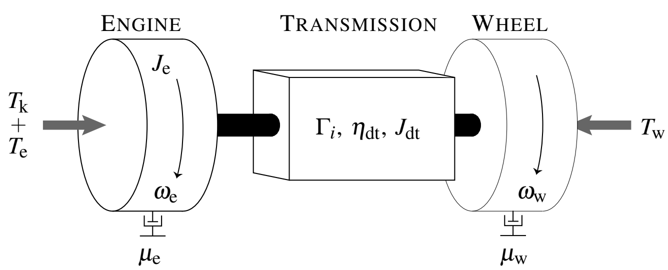

Figure 9: Model schematic of the 1-DOF system. As opposed to the previous phase, the layshaft and engine speeds are equal, i.e., $\omega_l = \omega_e$, and the clutch torque is now determined by a torque balance (hence not shown in the schematic). Also, the engine speed ω_e and the wheel speed ω_w are now directly linked through the gear ratio Γ_i. The other quantities are equal to the previous model.

Due to the clutch being locked, the wheel's rotational speed is determined by the engine speed and the gear ratio. In this phase it is therefore not treated as a dynamic state anymore:

$$\omega_w(t) = \omega_e(t) \cdot \Gamma_i(t), \qquad i \in \{1, \ldots n_{gears}\}. \tag{18}$$

As opposed to Phase 1, where we assumed that the system is always in first gear, the dynamics are now piecewise-defined based on the gear i. There is one set of differential equations for each gear i in Phase 2, which is equivalent to having multiple concatenated Phases 2, called Phase 2_i.

D. Longitudinal Dynamics

The longitudinal dynamics of the car are given by the linear momentum principle:

$$\frac{\mathrm{d}}{\mathrm{d}t} v_{car}(t) = \frac{1}{m_{car}} \cdot \Big(2 \cdot F_{trac}(t) - F_{drag}(t)\Big), \tag{19}$$

where F_{drag} is the aerodynamic drag force determined as

$$F_{drag}(t) = \frac{1}{2} \cdot c_d \cdot A_{car} \cdot \rho_{air} \cdot v_{car}(t)^2, \tag{20}$$

with c_d being the drag coefficient. Note that the left hand side of Equation (19) is also equal to the acceleration of the vehicle, i.e.,

$$\frac{\mathrm{d}}{\mathrm{d}t} v_{car}(t) = a_{car}(t). \tag{21}$$

As we are performing the optimisation in time domain, we introduce an additional state for the position s, which evolves according to

$$\frac{\mathrm{d}}{\mathrm{d}t} s(t) = v(t). \tag{22}$$

E. Energy Recovery System

In our high-level model, the Energy Recovery System (ERS) is composed of the MGU-K and the battery. The battery energy E_b evolves according to

$$\frac{\mathrm{d}}{\mathrm{d}t} E_b(t) = -P_b(t), \tag{23}$$

where P_b is the power depleting the battery, i.e.,

$$P_b(t) = P_{k,dc}(t) + \alpha_b \cdot P_{k,dc}(t)^2, \tag{24}$$

with $P_{k,dc}$ being the electric MGU-K power, which is modelled as

$$P_{k,dc}(t) = P_k(t) + \alpha_k \cdot P_k(t)^2. \tag{25}$$

where α_b and α_k are loss parameters that account for mechanical-to-electrical and electrical-to-mechanical losses. Finally, the mechanical MGU-K power can be computed as

$$P_k(t) = T_k(t) \cdot \omega_k(t),$$

where ω_k is the motor's rotational speed, which according to regulations is also constrained:

$$-60\,000\,\text{rpm} \quad \leqslant \quad \omega_k(t) \quad \leqslant \quad 60\,000\,\text{rpm}. \tag{26}$$

F. Fuel Energy

The final model concerns the fuel energy. We include this simple integrator in order to provide sensible comparisons in the case studies, i.e.,

$$\frac{\mathrm{d}}{\mathrm{d}t} E_f(t) = P_f(t), \tag{27}$$

where the fuel power P_f is defined as

$$P_f(t) = \frac{P_e(t)}{\eta_{therm}},$$

where η_{therm} is a constant thermal efficiency of the ICE, and

$$P_e(t) = T_e(t) \cdot \omega_e(t)$$

is the engine power.

III. Methodology

In this section, we formulate the OCP, whose solution delivers the optimal launch strategy, and discuss how to solve this problem numerically. In particular, we formulate a multi-phase OCP to account for the hybrid dynamics, the discrete changes between Phase 1 and 2, and the gear shifts. As the order of discrete changes is known a priori, there is no combinatorial aspect in this problem and we only need to optimise the durations of each phase. However, due to the dynamics' nonlinearity and the introduced time transformations, the problem is numerically challenging to solve. To cope with this, we take a direct approach to optimal control. That is, the continuous-time multi-phase OCP is discretised with the direct multiple shooting approach. To integrate the dynamics, we employ an explicit Runge Kutta 4 method. First, we devise the OCP for each of the Phases 1 and 2, then we discuss the joint optimisation problem. In each case we first define the differential states, control inputs, and other optimisation variables. Next, we define the cost function for the corresponding phase, and then we complete the OCP formulation using the dynamic model from the previous sections, and the corresponding constraints.

A. Phase 1

First, we define the OCP for optimising Phase 1. The differential states $\mathbf{x}_1$ of the dynamic model are

$$\begin{aligned}\mathbf{x}_1(t) = \big[v_{\text{car}}(t) \quad \vartheta_{\text{c}}(t) \quad \omega_{\text{e}}(t) \quad \omega_{\text{w}}(t) \quad \dots\\ \dots \quad E_{\text{b}}(t) \quad s(t) \quad E_{\text{f}}(t)\big]^\top,\end{aligned} \tag{28}$$

and evolve according to Equations (19), (9), (12), (16), (23), (22), and (27), respectively.

The control inputs $\mathbf{u}_1$ are

$$\mathbf{u}_1(t) = \big[r_{\text{c}}(t) \quad T_{\text{e}}(t) \quad T_{\text{k}}(t)\big]^\top. \tag{29}$$

An additional optimisation variable is the final time $t_{N,1}$, representing the duration of Phase 1.

As we aim to maximise vehicle acceleration, we propose the following cost function:

$$\begin{aligned}J_1(\mathbf{x}_1,\mathbf{u}_1,t_{N,1}) = \int_0^{t_{N_1}} \Big[\gamma_{J,1}\cdot\big(v_{\text{slip}}(\tau)-v_{\text{slip,opt}}\big)^2 + \dots\\ \gamma_{J,2}\cdot\big(\frac{\mathrm{d}}{\mathrm{d}t}T_{\text{tot}}(\tau)\big)^2\Big]\mathrm{d}\tau + \dots\\ \gamma_{J,3}\cdot t_{N,1}.\end{aligned}$$

In the stage cost, we penalise the square deviation from the optimal slip speed weighted by a tuning parameter $\gamma_{J,1}$. To highlight the clutch paddle optimisation, we also include a $\gamma_{J,2}$-weighted penalty on the squared time derivative of the total torque input, characterised as the sum of engine and MGU-K torque. This assumption stems from the fact that the driver mainly focuses on the clutch operation during Phase 1, rather than the throttle pedal, which is kept almost constant. In the terminal cost, we introduce a penalty on the final time of Phase 1 $t_{N,1}$ weighted by a tuning parameter $\gamma_{J,3}$, to achieve clutch locking, and thus the end of Phase 1, at lower engine speeds.

Problem 1. *The continuous-time OCP for Phase 1 reads*

$$\min_{\mathbf{x}_1(\cdot),\mathbf{u}_1(\cdot),t_{N,1}} \quad J_1(\mathbf{x}_1,\mathbf{u}_1,t_{N,1}),$$

subject to

$$\textit{Diff. Equations:} \quad (19),(9),(12),(16),(23),(22),(27),$$

and the following path constraints:

$$\begin{aligned} &\textit{Tyre:} && (2),(3)-(6),\\ &\textit{Clutch:} && (8),(10),\\ &\textit{Drivetrain:} && (13)\\ &\textit{Long. Dyn.:} && (20),(21),\\ &\textit{ERS:} && (24)-(26),\\ &\textit{Inputs:} && (7),(14),(15).\end{aligned}$$

The initial conditions of the OCP are

$$\begin{aligned}\mathbf{x}_1(0) = \big[0\,\text{km/h} \quad \vartheta_{\text{c,init}} \quad \omega_{\text{e,init}} \quad 0\,\text{rpm} \quad \dots\\ \dots \quad 4\,\text{MJ} \quad 0\,\text{m} \quad 0\,\text{MJ}\big]^\top,\end{aligned} \tag{30}$$

where $\vartheta_{\text{c,init}}$ and $\omega_{\text{e,init}}$ are tuning parameters. The terminal condition is characterised by the engine speed to be equal to the layshaft speed, i.e.,

$$\omega_{\text{e}}(t_{N,1}) = \frac{\omega_{\text{w}}(t_{N,1})}{\Gamma_1}. \tag{31}$$

B. Phase 2_i

This phase represents the locked clutch condition. We further subdivide Phase 2 into i additional phases, i.e., Phase 2_i, for each gear $i \in \{1,\dots,n_{\text{gears}}\}$. The differential states $\mathbf{x}_{2,i}$ now read

$$\begin{aligned}\mathbf{x}_{2,i}(t) = \big[v_{\text{car}}(t) \quad \vartheta_{\text{c}}(t) \quad \omega_{\text{e}}(t) \quad \dots\\ \dots \quad E_{\text{b}}(t) \quad s(t) \quad E_{\text{f}}(t)\big]^\top.\end{aligned} \tag{32}$$

The control inputs $\mathbf{u}_{2,i}$ are

$$\mathbf{u}_{2,i}(t) = \big[T_{\text{e}}(t) \quad T_{\text{k}}(t)\big]^\top. \tag{33}$$

Additional optimisation variables are the final times of each Phase $t_{N,2,i}$, each defining the length of the respective Phase 2_i.

Due to the same overall goal of maximising acceleration, the cost function still includes the square of the optimal deviation from slip speed, weighted with a tuning parameter $\gamma_{J,4}$, i.e.,

$$J_{2,i}(\mathbf{x}_{2,i},\mathbf{u}_{2,i},t_{N,2,i}) = \int_0^{t_{N,2,i}} \Big[\gamma_{J,4}\cdot\big(v_{\text{slip}}(\tau)-v_{\text{slip,opt}}\big)^2\Big]\mathrm{d}\tau.$$

Problem 2. *The continuous-time OCP for each Phase 2_i reads*

$$\min_{\mathbf{x}_{2,i}(\cdot),\mathbf{u}_{2,i}(\cdot),t_{N,2,i}} \quad J_{2,i}(\mathbf{x}_{2,i},\mathbf{u}_{2,i},t_{N,2,i}),$$

subject to

$$\textit{Diff. Equations:} \quad (19),(9),(17),(23),(22),(27),$$

and the following path constraints:

$$
\begin{aligned}
&\textit{Tyre:} && (2),(3)-(6),\\
&\textit{Clutch:} && (11),\\
&\textit{Drivetrain:} && (13),(18),\\
&\textit{Long. Dyn.:} && (20),(21),\\
&\textit{ERS:} && (24)-(26),\\
&\textit{Inputs:} && (14),(15).
\end{aligned}
$$

The initial conditions for Phase 2_i are set to the final conditions of the previous phase. The terminal conditions for Phase 2_i differ depending on the engaged gear. In particular, for all gears except for the last one, the terminal condition is that the engine speed needs to lie within a window in which shifting gear is feasible, i.e.,

$$\omega_{\mathrm{e,shift,min}} \leqslant \omega_{\mathrm{e}}(t_{N,2,i}) \leqslant \omega_{\mathrm{e,shift,max}}, \tag{34}$$

where $\omega_{\mathrm{e,shift,min}}$ and $\omega_{\mathrm{e,shift,max}}$ are tuning parameters. For the last engaged gear, which depends on the distance between the starting position and the first corner, hence on the chosen track, we instead include the following final conditions:

$$
\begin{aligned}
&s(t_{N,2,i}) = s_{\mathrm{final}},\\
&E_{\mathrm{b}}(t_{N,2,i}) \geqslant E_{\mathrm{b,init}} - \Delta E_{\mathrm{b}},\\
&E_{\mathrm{f}}(t_{N,2,i}) \leqslant \Delta E_{\mathrm{f}},
\end{aligned}
\tag{35}
$$

where $E_{\mathrm{b,init}} = 4\,\mathrm{MJ}$ according to (30).

C. Combined Problem

The final OCP we want to solve is the combined problem, including Phase 1 and all Phases 2_i. The combined differential states $\mathbf{x}_{\mathrm{tot}}$ read

$$\mathbf{x}_{\mathrm{tot}}(t) = \begin{bmatrix}\mathbf{x}_1^\top(t) & \mathbf{x}_{2,i}^\top(t) & \dots & \mathbf{x}_{2,n_{\mathrm{gears}}}^\top(t)\end{bmatrix}^\top. \tag{36}$$

To ensure continuity between the states of $\mathbf{x}_1$ and $\mathbf{x}_{2,i}$, we need to introduce additional constraints at the instants where phase switches occur. For the transition between Phase 1 and Phase 2_1, the switch happens at $t_{N,1}$, and the constraints are

$$\mathbf{0} = \begin{bmatrix}
v_{\mathrm{car},N_1}(t_{N,1}) - v_{\mathrm{car},N_{2,1}}(t_{N,1})\\
\vartheta_{\mathrm{c},N_1}(t_{N,1}) - \vartheta_{\mathrm{c},N_{2,1}}(t_{N,1})\\
\omega_{\mathrm{e},N_1}(t_{N,1}) - \omega_{\mathrm{e},N_{2,1}}(t_{N,1})\\
E_{\mathrm{b},N_1}(t_{N,1}) - E_{\mathrm{b},N_{2,1}}(t_{N,1})\\
s_{N_1}(t_{N,1}) - s_{N_{2,1}}(t_{N,1})\\
E_{\mathrm{f},N_1}(t_{N,1}) - E_{\mathrm{f},N_{2,1}}(t_{N,1})
\end{bmatrix}. \tag{37}$$

Additionally, also the acceleration of the car needs to be continuous, i.e.,

$$0 = a_{\mathrm{car},N_1}(t_{N,1}) - a_{\mathrm{car},N_{2,1}}(t_{N,1}). \tag{38}$$

Note that the fourth state of Phase 1, i.e., the wheel speed, ceases to be a dynamic state subject to optimisation in the subsequent phases, as the clutch being locked relates it directly to the engine speed via the gear ratio, as shown in Equation (18).

The transitions between Phase 2_i and 2_{i+1} occur at the time instants τ_i, i.e.,

$$\tau_i = t_{N,1} + \sum_{k=1}^{i} t_{N,2,k}, \qquad \forall i \in \{1,\dots,n_{\mathrm{gears}}-1\}. \tag{39}$$

Given that the durations of each phase are optimisation variables, the times τ_i are implicitly determined by solving the optimisation problem. The continuity constraints on the states at all those instants read as follows:

$$\mathbf{0} = \begin{bmatrix}
v_{\mathrm{car},N_{2,i}}(\tau_i) - v_{\mathrm{car},N_{2,i+1}}(\tau_i)\\
\vartheta_{\mathrm{c},N_{2,i}}(\tau_i) - \vartheta_{\mathrm{c},N_{2,i+1}}(\tau_i)\\
\omega_{\mathrm{e},N_{2,i}}(\tau_i) - \omega_{\mathrm{e},N_{2,i+1}}(\tau_i)\cdot\frac{\Gamma_{i+1}}{\Gamma_i}\\
E_{\mathrm{b},N_{2,i}}(\tau_i) - E_{\mathrm{b},N_{2,i+1}}(\tau_i)\\
s_{N_{2,i}}(\tau_i) - s_{N_{2,i+1}}(\tau_i)\\
E_{\mathrm{f},N_{2,i}}(\tau_i) - E_{\mathrm{f},N_{2,i+1}}(\tau_i)
\end{bmatrix}, \tag{40}$$

while for the acceleration we impose

$$0 = a_{\mathrm{car,N_{2,i}}}(\tau_i) - a_{\mathrm{car,N_{2,i+1}}}(\tau_i). \tag{41}$$

The combined control inputs $\mathbf{u}_{\mathrm{tot}}$ are

$$\mathbf{u}_{\mathrm{tot}}(t) = \begin{bmatrix}\mathbf{u}_1^\top(t) & \mathbf{u}_{2,i}^\top(t) & \dots & \mathbf{u}_{2,n_{\mathrm{gears}}}^\top(t)\end{bmatrix}^\top. \tag{42}$$

As in Sections III-A and III-B, also all the time durations of each phase, i.e., $t_{N,1}$, $t_{N,2,i}$ for each gear i, are optimisation variables. Finally, we recognise the state-dependency of Equation (15) and rewrite it to exploit the phase structure of the optimisation problem. In particular, we assume that the vehicle speed of $50\,\mathrm{km/h}$ is reached in Phase 2_1, i.e., in the clutch locked scenario in gear 1. We then further divide Phase 2_1 into two distinct phases, called Phase 2_{1a} and 2_{1b}. In the former, for all $t \in [0, t_{\mathrm{f}}]$, we impose

$$T_{\mathrm{k,min}}(\omega_{\mathrm{k}}(t)) \quad \leqslant \quad T_{\mathrm{k}}(t) \quad \leqslant \quad 0, \tag{43}$$

where t_{f} is the instant in which $v_{\mathrm{car}} = 50\,\mathrm{km/h}$. In the latter, for all $t \in [t_{\mathrm{f}}, t_{N,2,1}]$, we write

$$T_{\mathrm{k,min}}(\omega_{\mathrm{k}}(t)) \quad \leqslant \quad T_{\mathrm{k}}(t) \quad \leqslant \quad T_{\mathrm{k,max}}(\omega_{\mathrm{k}}(t)). \tag{44}$$

According to the previous definition of τ_i,

$$
\begin{aligned}
\tau_{1a} &= t_{N,1} + t_f,\\
\tau_{1b} &= t_{N,1} + t_{N,2,1}.
\end{aligned}
$$

The terminal condition of Phase 2_{1a} is

$$v_{\mathrm{car}}(\tau_{1a}) = 50\,\mathrm{km/h}. \tag{45}$$

This further division adds yet another set of continuity constraints to the time instant τ_{1a} for the states of the adjacent phases according to Equation (40) and (41).

Problem 3. *The combined continuous time OCP reads as follows:*

$$\min_{\substack{\mathbf{x}_1(\cdot),\mathbf{u}_1(\cdot),t_{N,1},\\ \mathbf{x}_{2,i}(\cdot),\mathbf{u}_{2,i}(\cdot),t_{N,2,i}}} \quad J_1(\mathbf{x}_1,\mathbf{u}_1) + \sum_{i=1}^{n_{\mathrm{gears}}} J_{2,i}(\mathbf{x}_{2,i},\mathbf{u}_{2,i}),$$

subject to the constraints of Problem 1 for Phase 1, the constraints of Problem 2 for all Phases 2_i, and the following additional gluing constraints:

$$
\begin{aligned}
&\textit{Phase } 1-2\textit{:} && (37),(38)\\
&\textit{Phase } 2_i - 2_{i+1}\textit{:} && (40),(41)\\
&\textit{Inputs Phase } 2_{1a}\textit{:} && (43)\\
&\textit{Inputs Phase } 2_{1b}\textit{:} && (44)
\end{aligned}
$$

D. Discretisation

To solve it numerically, the OCP is discretised into a nonlinear program (NLP) using a direct method, i.e., discretised in time via direct multiple shooting [51]. The step size Δt_1 of Phase 1 depends on the optimised final time $t_{N,1}$, and on the fixed number of discrete intervals N_1 (chosen a priori), i.e.,

$$\Delta t_1 = \frac{t_{N,1}}{N_1}. \tag{46}$$

Similarly, for Phases 2_i, the step size $\Delta t_{2,i}$ depends on the optimised final time $t_{N,2,i}$, and on the number of fixed discrete intervals $N_{2,i}$, i.e.,

$$\Delta t_{2,i} = \frac{t_{N,2,i}}{N_{2,i}}, \qquad \forall i \in \{1,\ldots,n_{\text{gears}}\}. \tag{47}$$

The amount of control intervals of the combined problem is the sum of all single ones, i.e.,

$$N = N_1 + N_{2,1} + \ldots + N_{2,n_{\text{gears}}}. \tag{48}$$

After the discretisation of the OCP in time, we obtain a NLP which we solve numerically with IPOPT via its CasADi interface [52], [53]. Readers not familiar with direct optimal control are referred to Chapter 8 of [54].

IV. Results

In this section we showcase two case studies representing real life scenarios that are likely to occur within a season. In particular, we analyse the optimal launch up to the first corner for the Circuit de Monaco (MON). The drivers starting from the front are the closest to the first corner, i.e.,

$$s_{\text{final}} = 140\,\text{m}.$$

According to telemetry data from previous seasons, the last gear that they engage on this circuit before reaching the braking point is gear 5. Accordingly, we can relate the terminal condition on the distance in (35) to that specific gear, i.e.,

$$s(\tau_4 + t_{N,5}) = s_{\text{final}}.$$

In the first case study we also comment on the optimal solution resulting from the separate optimisation, i.e., each phase is sequentially optimised. This validation highlights the advantage of the optimisation being accomplished jointly over the entire horizon.

A. Case Study I – Limiting Battery Energy

In a first case study, we analyse how the optimal launch varies depending on the imposed final energy battery target ΔE_b in (35). In particular, we consider an unconstrained case, resulting in $\Delta E_\text{b} = 0.9\,\text{MJ}$, and compare it to a constrained scenario with $\Delta E_\text{b} = 0.8\,\text{MJ}$. In real life, saving battery during the launch would optimally come at the expense of more fuel usage, as the importance to reach the corner as fast as possible outweighs the increased energy consumption. Nonetheless, to allow for a fair comparison, we impose the same fuel target ΔE_f on both optimisations. In Table I we summarise the obtained KPIs.

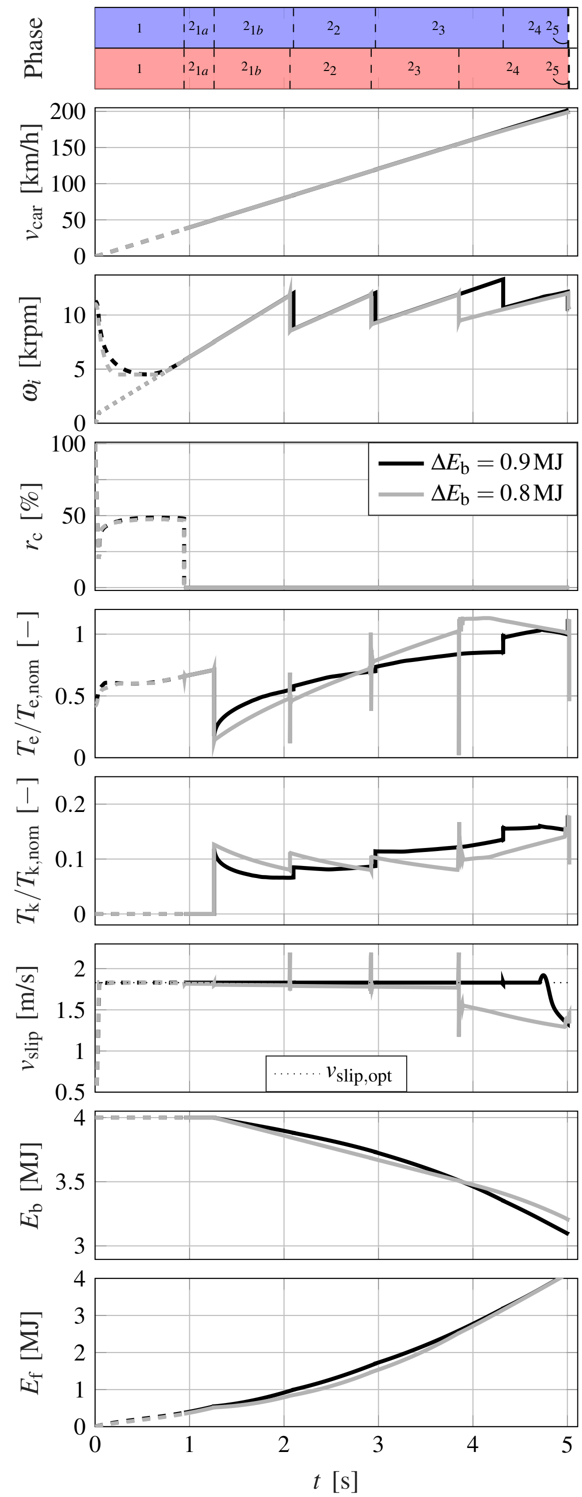

Figure 10: State and input trajectories for the case of unconstrained battery energy depletion and the one constrained to $\Delta E_\text{b} = 0.8\,\text{MJ}$. At the top we depict the various phases. In blue for the unconstrained, in red for the constrained case. The dashed interval represents Phase 1. Additionally, top-down, we show the vehicle's velocity, the rotational speed of engine and layshaft, the clutch input, the normalised engine and MGU-K torques, the slip speed, the battery energy, and the fuel energy. Please note that the clutch paddle input for Phase 2 is artificially set to 0, as the input itself does not exist anymore.

In Fig. 10 we display the state and input trajectories of the two scenarios. Above the top plot we depict the different phases for both scenarios, i.e., the unconstrained battery in blue, the constrained one in red. The dashed region represents Phase 1. The engine and layshaft speeds start at $\omega_{\mathrm{e,init}}$ and 0 rpm, respectively, and then get closer to each other until synchronisation. To achieve the latter, the clutch paddle shows a sudden release down to 25 %. This instant release is supposed to fast accelerate the inertias downstream. Nonetheless, the paddle's position is then increased to about 50 % to keep the engine speed constant, and ensure a smooth synchronisation. At the point where the two speeds meet, the clutch paddle is entirely released. Meanwhile, the engine torque merely increases slightly, as the main actuator in this phase is the clutch. Interestingly, the duration of Phase 1 for both scenarios is almost equal. This suggests that despite the energy limitation, a faster vehicle velocity increase at low speeds is more beneficial than saving energy for later use. Also, decreasing the duration of Phase 1 in which some energy is wasted into heat, is always optimal independently on targets.

The torque trajectories show the split between ICE and MGU-K torques. Given the limitation on both energy storages, we can observe that as soon as 50 km/h are reached, the MGU-K of the constrained scenario is operated more forcefully compared to the unconstrained one. This behaviour depletes the battery more while saving fuel. Also, we notice that gears 2 and 3 are engaged slightly earlier, while gear 4 almost 500 ms earlier. This anticipates the increase of MGU-K torque at the wheels to lower speeds, leading to a more efficient battery energy deployment in terms of acceleration performance. After around 2.8 s, this trend reverses, and the constrained ICE torque surpasses the unconstrained one upon reaching the same fuel target. At the same time, the constrained MGU-K torque falls below the unconstrained one, to meet also the battery target. In terms of slip speed, we show how during Phase 1 both scenarios are at the optimal value. In racing, we call this a grip-limited region, as the deployed torque by the PU is below the maximum torque. Specifically, the PU delivers only the torque that can be fully transferred to the track for propulsion. Once Phase 2 begins, the slip speed of the unconstrained scenario is at its optimum, and thus grip-limited, until the end of Phase 2_4. From then, the torque deployable by the PU is not sufficient to stay at the optimal slip speed, entering the so-called power-limited region. The constrained scenario leaves the grip-limited region steadily throughout the optimisation horizon, as the torque is reduced to save energy.

Finally, we comment on the differences between joint optimisation proposed in this framework and sequential phase optimisation. As soon as target constraints are introduced, the sequential optimisation is not able to guarantee a solution. In fact, for the constrained scenario proposed in this case study, the sequential optimisation becomes infeasible, as it is physically impossible to reach the imposed target in Phase 2_5 starting from the battery energy reached at the end of Phase 2_4. Only a careful tuning of each phase's terminal conditions would guarantee a solution, although it requires a priori knowledge of the solution or could potentially bias it. Hence, we do not include a graphical comparison.

ΔE_b	0.9 MJ		0.8 MJ	
	τ_i [s]	v_car [km/h]	τ_i [s]	v_car [km/h]
Phase 1	0.9424	37.13	0.9416	37.13
Phase $\mathbf{2}_{1a}$	1.2601	50	1.2594	50
Phase $\mathbf{2}_{1b}$	2.1002	84.04	2.0633	82.57
Phase $\mathbf{2}_2$	2.9678	119.19	2.9227	117.37
Phase $\mathbf{2}_3$	4.3201	173.99	3.8511	154.95
Phase $\mathbf{2}_4$	5.0040	201.28	5.0104	198.92
Phase $\mathbf{2}_5$	<u>5.0140</u>	<u>201.65</u>	<u>5.0204</u>	<u>199.30</u>

Table I: In this table we summarise the key performance indicators of the first case study. In particular, we look at the time instants of phase switches, and the velocity at which they occur. To assess the overall performance, we include the final time taken to reach the first braking point for the two scenarios.

B. Case Study II – Dry vs. Wet Conditions

In the second case study we analyse the influence of track conditions on the optimal launch strategy. Although under heavy raining conditions drivers sometimes prefer to start their race in second gear, in this example we still assume the launch phase to begin in first gear. In order to emulate lower grip due to track conditions, we adapt the friction curve as shown in Fig. 5. The maximum force that can be transferred by the tyre is decreased by 20 % and the optimal slip speed increases. Concerning the energy targets, we do not specify a target for the wet track. In particular, we also want to investigate how the energy management varies under more restricting environmental conditions. The dry track result considered is the unconstrained scenario of Section IV-A.

Fig. 11 shows the state and input trajectories for this case study. As in Section IV-A, above the top plot we depict the different phases for both scenarios, i.e., the dry condition in blue, the wet one in red. In the top plot we recognise the decreased velocity evolution of the wet scenario due to missing grip. As a consequence, the target position s_final is reached about 0.94 s later. In terms of Phase 1, we can observe that the duration of the clutch slipping phase for the wet track increases by about 25 %. This can be attributed to the driver releasing the clutch paddle more slowly, and thus transferring less torque through the clutch to maintain traction. The torque trajectories show that also for the subsequent phases the torque deployment is reduced compared to the dry scenario. Nonetheless, the slip speed trajectory shows that the grip-limited condition is never left. In particular, the optimal value is maintained, proving that the torque providing maximal propulsion is deployed. The dry scenario, as commented on above, enters the power-limited region towards the end of gear 4. Once the power-limited region is reached, full torque is deployed by the PU, as traction is not a limiting factor anymore.

As a consequence of the lower acceleration, the threshold of 50 km/h is reached about 0.5 s later, meaning that also the beginning of MGU-K usage is shifted to the back. Therefore, as summarised in Table II, the subsequent phases last longer for the wet track conditions, too, resulting in reaching the target 0.94 s later. Generally, given the decreased total torque that can be transferred, we notice that the battery energy is depleted 0.25 MJ less, whilst consuming 0.5 MJ less of fuel energy. Eventually, gear 5, and thus Phase 2_5, is not

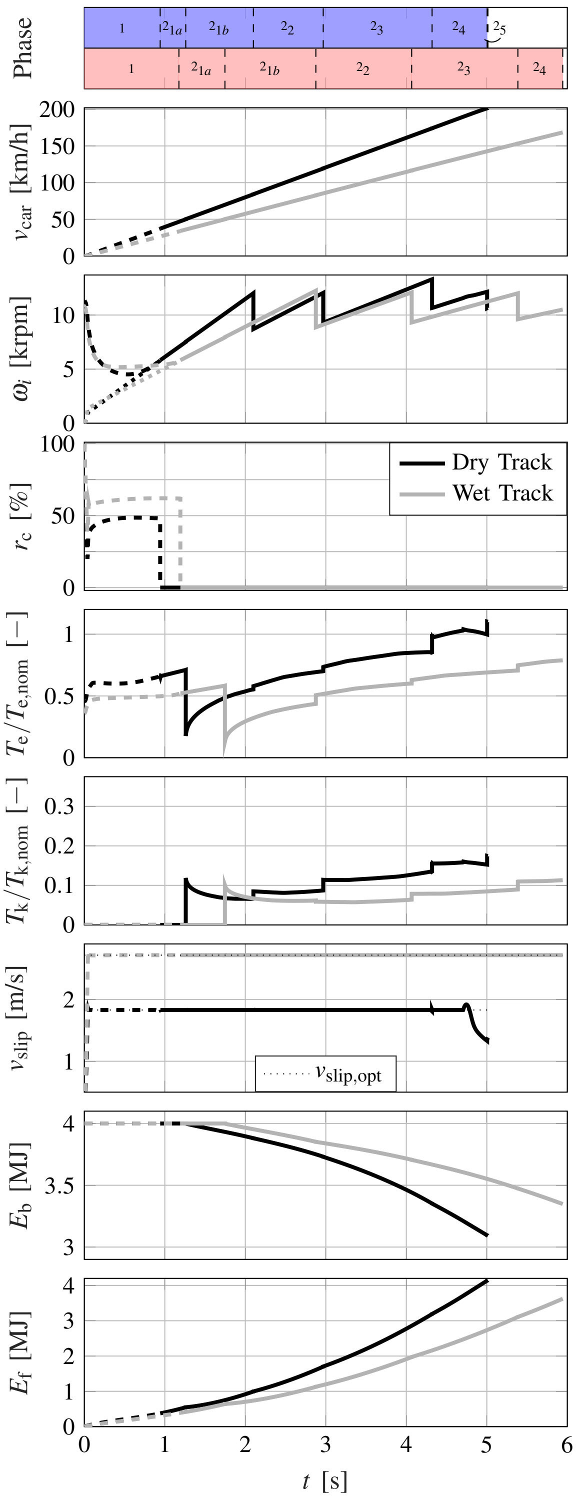

Figure 11: State and input trajectories for the case of dry track and wet track conditions. At the top we depict the various phases. In blue for the dry, in red for the wet case. The dashed interval represents Phase 1. Additionally, top-down, we show the vehicle's velocity, the rotational speed of engine and layshaft, the clutch input, the normalised engine and MGU-K torques, the slip speed, the battery energy, and the fuel energy. Please note that the clutch paddle input for Phase 2 is artificially set to 0, as the input itself does not exist anymore.

	Dry Track		Wet Track	
	τ_i [s]	v_{car} [km/h]	τ_i [s]	v_{car} [km/h]
Phase 1	0.9424	37.13	1.1932	33.94
Phase 2$_{1a}$	1.2601	50	1.7467	50
Phase 2$_{1b}$	2.1002	84.04	2.8763	82.56
Phase 2$_{2}$	2.9678	119.19	4.0667	116.42
Phase 2$_{3}$	4.3201	173.99	5.3858	153.18
Phase 2$_{4}$	5.0040	201.28	<u>5.9433</u>	<u>168.41</u>
Phase 2$_{5}$	<u>5.0140</u>	<u>201.65</u>	–	–

Table II: In this table we summarise the key performance indicators of the second case study. In particular, we look at the time instants of phase switches, and the velocity at which they occur. To assess the overall performance, we include the final time taken to reach the first braking point for the two scenarios. Note that in the wet scenario gear 5 is never reached.

even reached before the end of the straight, meaning that the terminal condition is

$$s(\tau_3 + t_{N,4}) = s_{\text{final}}.$$

V. Conclusion & Outlook

This research marks a significant milestone in the optimisation of launch procedures in motor sports. We have shown a novel framework, able to jointly optimise many phases that evolve according to different dynamics. The optimisation consists of finding the optimal high-level inputs to the hybrid electric power unit and the instants of phase-switching, i.e., from a slipping to a locked clutch and from any gear to the subsequent one. We have shown that while a sequential phase optimisation results in infeasibilities, the proposed framework is able to handle terminal constraints in the last phase. By means of two real-life scenarios we showed that the clutch actuation, and thus the duration of the clutch slipping phase, does not change with varying energy budgets, suggesting that a fast shaft speed synchronisation is always preferred. Additionally, we have demonstrated how a decrease in available battery energy is coped with through an alternative shifting strategy. Finally, we have assessed the influence of varying grip levels, i.e., wet track scenarios, on the optimal clutch paddle trajectory and torque deployment. Specifically, when a grip decrease of 20 % is considered, we have demonstrated a duration increase of the slipping phase by 25 %, and a total time increase of 0.94 s with over 30 km/h velocity difference.

Further development of this framework includes the extension of the model from a high-level perspective to a low-level one. This would allow for the inclusion of the phase prior to the red lights turning off. In particular, the power unit preparation during standstill could be analysed. Furthermore, this framework could be augmented with an additional optimisation problem attached to the end of the launch horizon. Exploiting the same method presented in this work of combining multiple phases, the new optimisation problem would consider the remainder of the lap. This would allow the determination of the optimal energy targets that need to be achieved at the first braking point, from a lap time perspective.

ACKNOWLEDGEMENT

We would like to thank Ferrari S.p.A. for supporting this project. Moreover, we would like to express our gratitude to Dr. Ilse New for her helpful and valuable comments during the proofreading phase.

REFERENCES

[1] FIA, "2024 Formula one technical regulations," FIA, Geneva, Switzerland, Tech. Rep., 2024.

[2] ——, "2024 Formula one sporting regulations," FIA, Geneva, Switzerland, Tech. Rep., 2024.

[3] McLaren. (2020) Lights out and away we go? [Online]. Available: https://www.mclaren.com/racing/team/carlos-sainz-lando-norris-reaction-time/

[4] Ergast. (2024) Ergast API - a public open source formula one API. [Online]. Available: http://ergast.com/mrd/

[5] L. Guzzella and A. Sciarretta, *Vehicle Propulsion Systems: Introduction to Modeling and Optimization*, 3rd ed. Berlin, Heidelberg: Springer, 2013.

[6] W. F. Milliken and D. L. Milliken, *Race Car Vehicle Dynamics*, 1st ed. Warrendale, Pennsylvania: Society of Automotive Engineers, 1995.

[7] R. N. Jazar, *Vehicle Dynamics: Theory and Application*, 1st ed. New York, NY: Springer Science + Business Media, 2008.

[8] J. D. Setiawan, M. Safarudin, and A. Singh, "Modeling, simulation and validation of 14 dof full vehicle model," in *Proc. ICICI-BME*. Semarang, Indonesia; Ayer Keroh, Malaysia: IEEE, 2009.

[9] R. Isermann, "Vehicle dynamics modeling," in *Automotive Control*, 1st ed. Springer, Berlin, Heidelberg, 2022, ch. 2, pp. 15–35.

[10] R. Höglund, "Modelling of a high-performance vehicle in matlab/simulink and canopy simulations," Master's thesis, KTH Royal Institute of Technology, Stockholm, Sweden, 2022. [Online]. Available: https://www.essays.se/essay/2c15493802/

[11] J. Wong, "Longitudinal vehicle dynamics," in *Theory of Ground Vehicles*, 3rd ed. Hoboken, NJ: Wiley-Interscience, 2001, pp. 96–119.

[12] A. Serrarens, M. Dassen, and M. Steinbuch, "Simulation and control of an automotive dry clutch," in *Proceedings of the 2004 ACC*, vol. 5, 2004, pp. 4078–4083.

[13] A. Chintala, R. L. Narayana, and C. Srinivas, "Modelling, simulation and control of an automotive clutch system," *International Journal of Engineering Research & Technology (IJERT)*, vol. 2, no. 12, pp. 3422–3427, 2013.

[14] H. N. Quang, "Dynamic modelling of friction clutches and application of this model in simulation of drive systems," *Periodica Polytechnica Mechanical Engineering*, vol. 42, no. 2, pp. 169–181, 1998.

[15] G. Lawrence, "Friction characteristic prediction within a carbon-carbon race clutch," Ph.D. dissertation, Oxford Brookes University, 2008.

[16] R. Fono-Tamo, "A mathematical model for the purpose of analysing the thermal stress characteristics of pks-based brake pad with matlab," in *Materials Today: Proceedings*, vol. 5, 2018, pp. 12 534–12 544.

[17] A. Myklebust and L. Eriksson, "Torque model with fast and slow temperature dynamics of a slipping dry clutch," in *Proceedings of the 2012 IEEE Vehicle Power and Propulsion Conference (VPPC)*. Seoul, Korea: IEEE, 2012, pp. 851–856.

[18] T. Gkinis, R. Rahmani, H. Rahnejat, and M. O'Mahony, "Heat generation and transfer in automotive dry clutch engagement," *Journal of Zhejiang University-SCIENCE A*, vol. 19, no. 3, pp. 175–188, 2018.

[19] A. Crowther, N. Zhang, D. K. Liu, and J. K. Jeyakumaran, "Analysis and simulation of clutch engagement judder and stick-slip in automotive powertrain systems," *Proceedings of the Institution of Mechanical Engineers, Part D: Journal of Automobile Engineering*, vol. 218, no. 12, pp. 1427–1443, 2004.

[20] M. Zaplata, K. Grudzinski, and B. Warda, "Modelling and computer analysing the stick-slip motion in a mass-spring system with friction," in *Contact Mechanics II*, ser. Transactions on Engineering Sciences, vol. 7. Southampton, UK: WIT Press, 1995, pp. 488–494.

[21] F. Marín, F. Alhama, and J. Moreno, "Modelling of stick-slip behaviour with different hypotheses on friction forces," *International Journal of Engineering Science*, vol. 60, pp. 13–24, 2012.

[22] G. Pica, C. Cervone, A. Senatore, M. Lupo, and F. Vasca, "Dry dual clutch torque model with temperature and slip speed effects," *Intelligent Industrial Systems*, vol. 2, pp. 133–147, 2016.

[23] P. Dolcini, C. C. de Wit, and H. Béchart, "Improved optimal control of dry clutch engagement," in *Proceedings of the 16th IFAC World Congress*. Prague, Czech Republic: IFAC, 2005, pp. 25–30.

[24] F. Garofalo, L. Glielmo, L. Iannelli, and F. Vasca, "Smooth engagement for automotive dry clutch," in *Proceedings of the 40th IEEE Conference on Decision and Control*. Orlando, Florida, USA: IEEE, 2001, pp. 3447–3452.

[25] D. Anderson, *Brush Tire Model for Vehicle Dynamics*, 1st ed. Oxford, UK: Elsevier Inc., 2018.

[26] R. Krishnakumar, "Brush tire model for vehicle dynamics," 2021, lecture materials from the Vehicle Dynamics course. [Online]. Available: https://archive.nptel.ac.in

[27] H. B. Pacejka, *Tyre and Vehicle Dynamics*, 2nd ed. Oxford, UK: Butterworth-Heinemann, Elsevier, 2006.

[28] F. Tarak, "Pacejka tire model: Magic formula," 2021, lecture notes on the application of the Magic Formula in tire modeling using MATLAB/Simulink. [Online]. Available: http://www.racer.nl/reference/pacejka.htm

[29] J. E. Bernard and C. L. Clover, "Tire modeling for low-speed and high-speed calculations," in *Proceedings of the SAE International Congress and Exposition*. Detroit, Michigan, USA: SAE International, 1995.

[30] J. Hooker, "Optimal driving for single-vehicle fuel economy," *Transportation Research Part A: General*, vol. 22, no. 3, pp. 183–201, 1988.

[31] L. V. Pérez and E. A. Pilotta, "Optimal power split in a hybrid electric vehicle using direct transcription of an optimal control problem," *Mathematics and Computers in Simulation*, vol. 79, no. 6, pp. 1959–1970, 2009.

[32] G. Heppeler, M. Sonntag, and O. Sawodny, "Fuel efficiency analysis for simultaneous optimization of the velocity trajectory and the energy management in hybrid electric vehicles," *IFAC Proceedings Volumes*, vol. 47, no. 3, pp. 6612–6617, 2014.

[33] T. Nüesch, P. Elbert, M. Flankl, C. Onder, and L. Guzzella, "Convex optimization for the energy management of hybrid electric vehicles considering engine start and gearshift costs," *Energies*, vol. 7, no. 2, pp. 834–856, 2014.

[34] A. Sciarretta, M. Back, and L. Guzzella, "Optimal control of parallel hybrid electric vehicles," *IEEE Transactions on Control Systems Technology*, vol. 12, no. 3, pp. 352–363, 2004.

[35] A. Sciarretta and L. Guzzella, "Control of hybrid electric vehicles," *IEEE Control Systems Magazine*, vol. 27, no. 2, pp. 60–70, 2007.

[36] N. Kim, S. Cha, and H. Peng, "Optimal control of hybrid electric vehicles based on pontryagin's minimum principle," *IEEE Transactions on Control Systems Technologies*, vol. 19, no. 5, pp. 1279–1287, 2011.

[37] A. Sciarretta, G. De Nunzio, and L. L. Ojeda, "Optimal ecodriving control: Energy-efficient driving of road vehicles as an optimal control problem," *IEEE Control Systems Magazine*, vol. 35, no. 5, pp. 71–90, 2015.

[38] N. Robuschi, M. Salazar, P. Duhr, F. Braghin, and C. H. Onder, "Minimum-fuel engine on/off control for the energy management of hybrid electric vehicles via iterative linear programming," in *IFAC Symposium on Advances in Automotive Control (AAC)*, 2019.

[39] J. Ritzmann, A. Christon, M. Salazar, and C. H. Onder, "Fuel-optimal power split and gear selection strategies for a hybrid electric vehicle," in *SAE Technical Paper 2019-24-0205*, 2019.

[40] C. Balerna, N. Lanzetti, M. Salazar, A. Cerofolini, and C. Onder, "Optimal low-level control strategies for a high-performance hybrid electric power unit," *Applied Energy*, vol. 276, 2020, Art. no. 115248.

[41] M. Veneri and M. Massaro, "A free-trajectory quasi-steady-state optimal-control method for minimum lap-time of race vehicles," *Vehicle System Dynamics*, vol. 57, no. 4, pp. 537–556, 2019.

[42] D. Limebeer, G. Perantoni, and A. Rao, "Optimal control of formula one car energy recovery systems," *International Journal of Control*, vol. 87, no. 10, pp. 2065–2080, 2014.

[43] M. Salazar, P. Elbert, S. Ebbesen, C. Bussi, and C. H. Onder, "Time-optimal control policy for a hybrid electric race car," *IEEE Transactions on Control Systems Technology*, vol. 25, no. 6, pp. 1921–1934, 2017.

[44] S. Ebbesen, M. Salazar, P. Elbert, C. Bussi, and C. H. Onder, "Time-optimal control strategies for a hybrid electric race car," *IEEE Transactions on Control Systems Technology*, vol. 26, no. 1, pp. 233–247, 2018.

[45] C. Balerna, M.-P. Neumann, N. Robuschi, P. Duhr, A. Cerofolini, V. Ravaglioli, and C. Onder, "Time-optimal low-level control and gearshift strategies for the formula 1 hybrid electric powertrain," *Energies*, vol. 14, no. 1, p. 171, 2021.

[46] M.-P. Neumann, G. Fieni, C. Balerna, P. Duhr, A. Cerofolini, and C. H. Onder, "Low-level online control of the formula 1 power unit with feedforward cylinder deactivation," *IEEE Transactions on Vehicular Technologies*, vol. 72, no. 7, pp. 8382–8397, 2023.

[47] A. Nurkanović, S. Albrecht, B. Brogliato, and M. Diehl, "The time-freezing reformulation for numerical optimal control of complementarity lagrangian systems with state jumps," *Automatica*, vol. 158, p. 111295, 2023.

[48] A. Nurkanović, T. Sartor, S. Albrecht, and M. Diehl, "A time-freezing approach for numerical optimal control of nonsmooth differential equations with state jumps," *IEEE Control Systems Letters*, vol. 5, no. 2, pp. 439–444, 2021.

[49] A. Nurkanović, J. Frey, A. Pozharskiy, and M. Diehl, "Finite elements with switch detection for numerical optimal control of rigid bodies with impacts and coulomb friction," *Nonlinear Analysis: Hybrid Systems*, vol. 52, p. 101460, 2024.

[50] A. Nurkanović and M. Diehl, "Continuous optimization for control of hybrid systems with hysteresis via time-freezing," *IEEE Control Systems Letters*, vol. 6, pp. 3182–3187, 2022.

[51] H. Bock and K. Plitt, "A multiple shooting algorithm for direct solution of optimal control problems*," *IFAC Proceedings Volumes*, vol. 17, no. 2, pp. 1603–1608, 1984, 9th IFAC World Congress, Budapest, Hungary, 2-6 July 1984.

[52] A. Wächter and L. Biegler, "Ipopt-an interior point optimizer," 2009.

[53] J. A. E. Andersson, J. Gillis, G. Horn, J. B. Rawlings, and M. Diehl, "CasADi – A software framework for nonlinear optimization and optimal control," *Mathematical Programming Computation*, vol. 11, no. 1, pp. 1–36, 2019.

[54] J. Rawlings, D. Mayne, and M. Diehl, *Model Predictive Control: Theory, Computation, and Design*. Nob Hill Publishing, 2017.